

Designing diversified crop rotations to advance sustainability: A method and an application

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ABSTRACT

Diversified crop rotations are considered key to solving sustainability issues with modern farming but trade-offs often occur among different sustainability objectives. It remains a challenge to design rotations that reach compromises across multiple sustainability domains. Here we propose a methodological framework for systematic design of compromise rotations in terms of a set of sustainability indicators, which for the first time combines a rotation generator model (ROTAT), Pareto-based decision-aid tools, and a multi-criteria characterization process. We exemplify this framework by exploring promising diversified rotations in terms of 11 sustainability indicators for Quzhou, a typical cereal-growing county in the North China Plain. ROTAT generated 3011 agronomically feasible rotations but none of them outperformed the currently dominant wheat-maize system in all indicators. Then 125 rotations (4 %) were selected as compromise rotations using the Pareto-based decision-aid methods. Finally, these compromise rotations were divided into eight clusters representing contrasting sustainability patterns, including two profitability-oriented, two nutrition-oriented, and two environment-oriented clusters, as well as two clusters that achieved balance across sustainability domains. We conclude that no single alternative rotation can solve all conflicting sustainability issues, but compromise alternatives could serve diverse stakeholder demands and provide well-planned options for future agricultural landscapes. The crop rotation design method showcased here could also be used in other specialized agricultural regions to inform stakeholders and policy-makers about possibilities and trade-offs when improving sustainability of crop production.

1. Introduction

Sustainable crop production is expected to be agronomically productive as well as environmentally sound and socio-economically viable (Deytieux et al., 2016). Since the Green Revolution, crop production has been highly intensified and specialized in many agricultural regions around the world (Ickowitz et al., 2019) so that a few calorie-dense cereal species dominate farmlands in short rotations with high resource inputs such as fertilizers, pesticides, and freshwater. In the past decades, such agricultural intensification and specialization has contributed to satisfying the dietary energy demand of the increasing global population but it has also caused negative environmental consequences (Foley et al., 2011; Pellegrini and Fernández, 2018), for

instance, water overexploitation and eutrophication (Withers et al., 2014; Pei et al., 2015), greenhouse gas (GHG) emissions (Bennetzen et al., 2016), and biodiversity loss (Beckmann et al., 2019). Moreover, with the pursuit of economic well-being and nutritional security, cereal-dominated agriculture offers limited scope for increased farm profit and micro-nutrient production (Garibaldi et al., 2017). All of these socio-economic, human-nutrition, and eco-environmental sustainability issues need to be addressed with equal importance and urgency. Therefore, it is crucial to redesign the current intensified and specialized cropping systems to advance agricultural sustainability.

Well-planned crop rotations, particularly the longer and more species-diverse ones, are thought to enhance agricultural sustainability for specialized cropping regions. Crop rotation is used to break the life

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cycle of pests and to improve soil fertility and resource use efficiency (Ouda et al., 2018). In modern agriculture, it further helps to handle socio-economic, human-nutrition, and environmental challenges. From a socio-economic perspective, crop rotation can enhance financial profitability by introducing high-value crops (Sánchez et al., 2022), and it can attenuate economic risks by combining crops with different temporal patterns of price volatility (Rosa-Schleich et al., 2019). From a nutritional perspective, diversifying crop rotations with pulses or vegetables could increase the production of dietary micronutrients such as minerals and vitamins to combat hidden hunger (Dwivedi et al., 2017). Diversified crop rotations may also serve to solve resource and environmental concerns. Previous studies reported that rotations with water-use-efficient species and a decreased cropping intensity could mitigate groundwater table decline (Yang et al., 2021). Furthermore, low-input rotations, for instance with legumes, reduce the reliance on synthetic fertilizers, thereby lowering associated pollution and GHG emission (Liu et al., 2022). Additionally, diversified crop rotations could be used as a measure to revive farmland biodiversity (Sietz et al., 2022). However, when multiple sustainability objectives are concurrently considered, trade-offs among objectives are often observed (Kanter et al., 2018), which increases the complexity and difficulty of crop rotation planning. In this situation, it is preferable to implement diversified crop rotations with compromises across sustainability domains while catering to various sustainability demands. However, there is a lack of a method to systematically design such promising rotations.

The North China Plain is one of the major specialized and intensified crop production regions in China and around the world. Here, the intensive wheat-maize double cropping system covers approximately 60 % of arable land area (Li and Lei, 2021). The widespread cultivation of the wheat-maize system has given rise to issues including low farm income, groundwater table decline, high nutrient losses, and pesticide risks to human and ecosystem health (Zhang et al., 2016; Wang et al., 2019; Mu et al., 2022). Moreover, local policies emphasize food and nutritional security, GHG emission abatement, and biodiversity conservation (MARA, 2022a). Thus, this region serves as an example of the arable farming areas that need to address multiple sustainability issues resulting from crop specialization while fulfilling diverse sustainability demands from agricultural policies and stakeholders. Since the wheat-maize system in itself cannot handle all of these challenges, the North China Plain presents an ideal case study area to explore diversified rotations that can achieve compromises across multiple sustainability objectives. Several previous studies evaluated a few expert-designed alternative rotations in the field trials but accounted for only a few sustainability indicators, e.g., grain and economic output, water consumption, and carbon emission (Zhao et al., 2021; Xiao et al., 2022), and overlooked others and potential trade-offs. So far, there have been limited efforts on systematic design of rotations with compromises across a comprehensive set of sustainability indicators for a given region.

The main objective of this study is to present a novel methodological framework for designing promising diversified rotations in terms of multiple sustainability attributes, which for the first time combines the rotation generator model with the multi-criteria decision-aid and characterization processes. Then, taking Quzhou, a typical cereal-dominated agricultural county in the North China Plain, as a case study area, we illustrated this framework to 1) design the rotations that represent compromises across the entire indicator set, and 2) systematically characterize those compromise rotations to understand their potential contribution to multi-attribute sustainability. These compromise rotations serve as the basis for local stakeholders' discussion on enhancing farming sustainability. This illustration shows how the presented method can be applied in a given area, thus helping to facilitate the broader application of this method to design well-performing rotations in any other specialized cropping regions that encounter various sustainability challenges.

2. Literature review

2.1. Model-based crop rotation generation and evaluation

Multiple crop rotation generator models have been devised to explore alternative rotations. This type of model generates all agronomically feasible rotations based on a list of candidate crops and a set of agronomic rules (Dury et al., 2012; Hernández-Ochoa et al., 2022). The agronomic, economic, and environmental performance of the generated rotations can be assessed ex-ante in an empirical way. For example, the ROTAT model was used to design rotations that enhance farm income and soil organic matter accumulation while lowering soil erosion for the vegetable-pasture farms in Uruguay (Dogliotti et al., 2004, 2014). Bachinger and Zander (2007) developed the ROTOR model to generate desirable rotations for organic farming systems in terms of operating profits, soil N balance, and weed infestation risks. Reckling et al. (2016) proposed a methodological framework for designing and evaluating feasible rotations in two European regions, highlighting the benefits of legume-inclusive rotations in economic gains, N emission mitigation, and phytosanitary effects. Compared with experience-based rotation design, the model-based design has three advantages: 1) it uses an explicit, systematic, and reproducible procedure, 2) it assesses to what extent each sustainability domain may benefit from improved rotations, and 3) it allows the analysis of synergies and trade-offs among sustainability domains (Dury et al., 2012; Carof et al., 2013).

However, rotation generator models are likely to produce large sets of feasible alternatives, which may overwhelm stakeholders and complicate decision-making, particularly when there are trade-offs among assessment indicators. Previous studies with rotation generator models often resort to thresholds for certain indicators to identify rotations for adoption (Dogliotti et al., 2004; Bachinger and Zander et al., 2007), but these threshold values are often arbitrarily defined and may not be available for all indicators. Furthermore, trade-offs among some indicators are overlooked in these studies. Hence it is crucial to integrate rotation generator models with threshold-free multi-criteria decision-aid tools to efficiently select well-performing solutions. Nevertheless, this integration remains under investigated yet.

2.2. Multi-criteria decision-aid methods on cropping systems

Previous studies on the ex-post assessment of cropping systems have often harnessed various quantitative decision-aid tools to choose desirable options in terms of multiple sustainability indicators (Deytieu et al., 2016; Ciccù et al., 2022). This is because, in most cases, there is no single system that performs best in all indicators. Most of these studies aggregate multiple elementary indicators into a composite one and recommend the crop sequence and/or management strategy with the highest overall sustainability score. This aggregation can be accomplished by calculating the additive sum or weighted average of normalized values of each indicator (Talukder et al., 2018; Ben Abdallah et al., 2022) or by quantifying the production efficiency such as data envelopment analysis (Pokhrel and Soni, 2017; Yang et al., 2022), eco-efficiency (Saber et al., 2021), and environmental footprints (Ankathi et al., 2019; Xian et al., 2023). While these methods can identify the 'best' cropping system based on the composite indicator, they may conceal the potential trade-offs among elementary indicators during the aggregation process. Therefore, the derived single optimal system perhaps cannot fit all kinds of sustainability demands.

Pareto-based decision-aid methods offer promise in addressing the aforementioned issue and thus have garnered increasing attention in multi-criteria assessment studies regarding agricultural systems (Adelhart Toorop et al., 2020; Liang et al., 2022; Ulukan et al., 2022). These methods use Pareto-based metrics to select the superior systems in the multi-criteria term. For instance, Liang et al. (2022) used the Pareto rank metric with seven socio-economic and eco-environmental indicators to select Pareto-optimal systems among a group of wheat-maize farmers in

the North China Plain. In another study, Groot et al. (2010) employed the Pareto efficiency metric to design agricultural landscapes that achieved better compromises across seven objectives related to economics, environment, and landscape characters. These methods take into account trade-offs among indicators during the selection process and ultimately maintain a diversity of promising options catering to different priorities. However, such metrics have not yet been utilized to support the systematic design of crop rotations that reconcile different sustainability domains.

3. Methods

3.1. Methodological framework

The present methodological framework contains three steps to systematically explore alternative crop rotations that reach compromises across multiple sustainability attributes for a given agricultural region (Fig. 1).

3.1.1. STEP1: Crop rotation generation & evaluation

All agronomically feasible rotations were generated by the ROTAT model using a list of locally cultivated crops and a set of rotation rules (Section 3.3). Simultaneously, ROTAT evaluated each rotation's sustainability performance based on crop input-output data collected from farmers' survey (Section 3.4).

3.1.2. STEP2: Crop rotation selection

This step helped to select compromise rotations across all indicators from the set of generated rotations (Section 3.5). Based on the indicator performance, we identified Pareto-optimal rotations using the Pareto ranking procedure and then filtered out Pareto-optimal rotations that had unacceptable trade-offs across indicators using Pareto efficiency score ordination.

3.1.3. STEP3: Crop rotation characterization

We finally applied a clustering technique to divide the whole set of compromise rotations into several clusters with contrasting sustainability patterns (Section 3.6). The outcome of this step would inform stakeholders to choose rotations tailored to given sustainability objectives.

3.2. Case study area

We applied the proposed methodological framework to design compromise rotations for Quzhou County, Hebei Province, China (114°50'E–115°13'E, 36°35'N–36°57'N, Fig. S1), which is a representative agricultural county in the North China Plain. Quzhou has a temperate continental monsoon climate with four distinct seasons. The mean annual temperature is 13.1 °C, and the mean annual precipitation is 500 mm, 60 % of which occurs from July to August. The soil type is calcareous alluvial, with a loamy texture, a pH of 7.5, and a soil organic matter content of 1.38 % (Li et al., 2022). Crop production in Quzhou is highly specialized and intensified because the winter wheat-summer

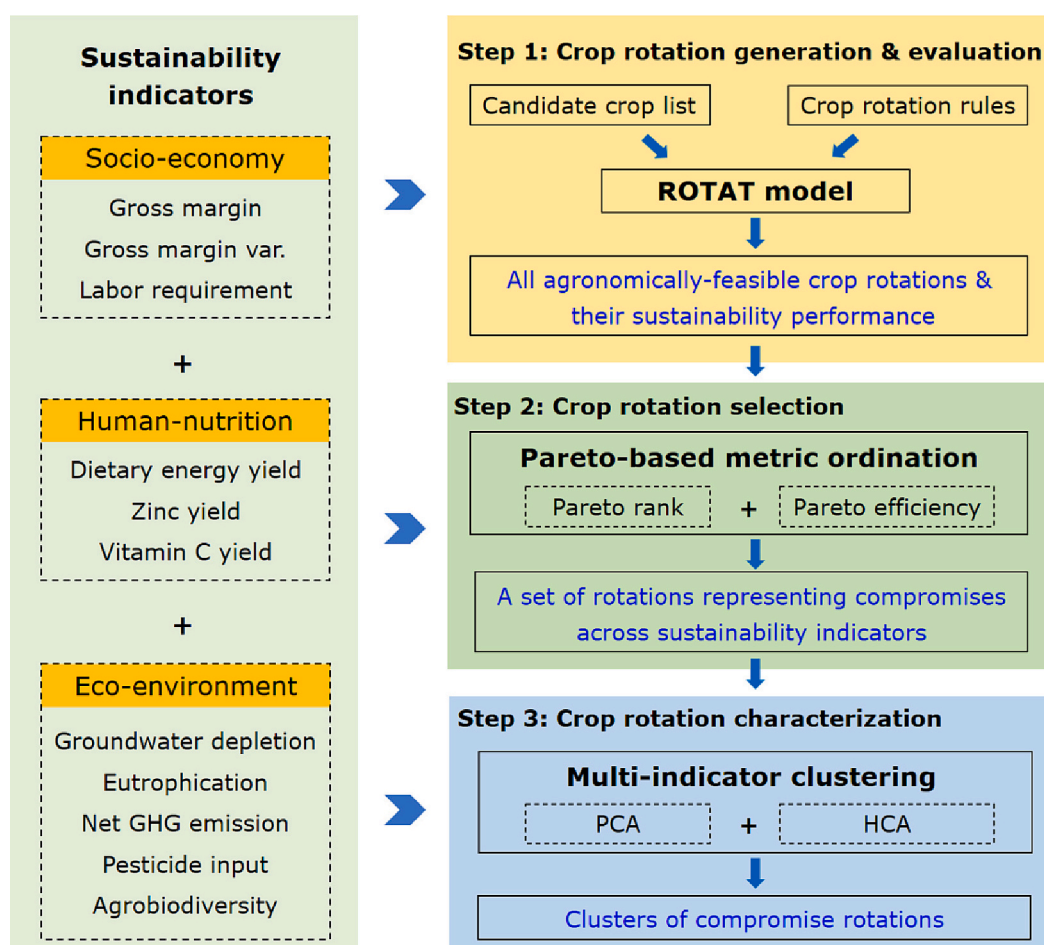


Fig. 1. The three-step methodological framework about the systematic exploration of crop rotation set representing compromises across multiple sustainability domains. PCA and HCA are the principal component analysis and the hierarchical clustering analysis respectively. Var. means variability. GHG means greenhouse gas.

maize double cropping system dominates local croplands, to which farmers apply considerable quantities of fertilizers and water (Liang et al., 2022). On the other hand, the pedo-climatic conditions are favorable for cultivation of various non-cereal crops such as grain legumes, cotton, stevia, and vegetables. Currently, this county is one of the pilot areas of The National Agriculture Green Development Strategy in China (MARA, 2022b), emphasizing the agricultural transition towards socio-economic, human-nutrition, and eco-environmental sustainability.

3.3. Crop rotation generation

We used the ROTAT model to explore alternative crop rotations for Quzhou based on ten local major crops (Section 3.3.2) and expert-defined agronomic rules (Section 3.3.3).

3.3.1. Model description

ROTAT is a static and rule-based model that generates agronomically feasible rotations at the field scale from a list of candidate crops (Dogliotti et al., 2003). The feasibility of generated rotations is ensured by obeying a set of pre-defined agronomic rules, which remove unfeasible rotations from the full factorial set of candidate crop combinations. These rules include crop-timing constraints (e.g., sowing and harvesting dates), sequence and frequency constraints (e.g., disallowed crop successions and the maximum frequency of each crop or crop group), and context-specific constraints (e.g., minimum and maximum rotation length). With such explicit rules, ROTAT-aided rotation generation is viewed as an objective and transparent process.

3.3.2. Candidate crops and their parameterization

To determine the list of candidate crops, we conducted a county-wide cropping system survey with 525 randomly selected farmers in 2020 and collected agronomic and input-output information on each crop that farmers cultivated (Liang et al., 2022; Supplementary Survey Questionnaire). We chose the ten most widely cultivated crop species for consideration, whereby the same crop species grown in a different growing season was considered a different crop (Table 1; Table S1). There were three candidate maize crops and two candidate foxtail millet crops, respectively, due to differences in growing season. We parameterized these crops in ROTAT based on the agronomic information and an inventory of inputs and outputs of each crop, which corresponded to the mean or mode values of observations from the survey. We used the average input-output data to ensure an accurate representation of the baseline performance of each candidate crop and avoid both over- and under-estimation. Using the survey information ensured that these candidate crops can be broadly grown in the focus region and reflect the

locally proven input-output relations.

3.3.3. Agronomic rules of crop rotation

By consulting with multiple local agronomists, we defined rules for the maximum frequency of crops and crop groups, the minimum period between repeated cultivation of crops and crop groups (Table 1), disallowed crop successions (Table S2), and the minimum and maximum rotation length. We set the minimum rotation length as one year and the maximum as four years because longer rotations are less practical for local farmers. The meaning of agronomic rules used in ROTAT is elaborated in Table S3. Following the rules would prevent abiotic and biotic stresses leading to substantial yield loss and it would also avoid long fallow periods between two successive crops for a rotation. Rotational constraints in this study were looser than those in other studies (Dogliotti et al., 2004; Reckling et al., 2016). For instance, maize and wheat were allowed to be planted every year and continuous cultivation of cotton was accepted, to reflect existing farmer practices because these crops have limited susceptibility to stresses induced by high-frequency cropping at current input levels according to local observations.

3.4. Crop rotation evaluation

We chose 11 indicators relevant to local sustainability challenges to comprehensively assess the performance of crop rotations by ROTAT. These indicators fell into the socio-economic, human-nutrition, and eco-environmental domains.

3.4.1. Model assumptions

The empirical assessment in ROTAT was based on crop input-output data representing locally common crop management. We found that farmers in Quzhou used high rates of inputs for candidate crops (Table S1). In this context of intensive management, we did not consider the effects of interactions between successive crops on crop yield in a rotation because high use of agricultural inputs, e.g., fertilizers, pesticides, and water, would mask the potential effects of preceding crops on nutrient and water turnover, and the prevalence of diseases, pests, and weeds (Zhao et al., 2020; MacLaren et al., 2022). For example, the crop yield of a certain crop species would be a fixed value independent of the preceding crop species in this study. This assumption was agreed upon by local agronomists according to their observations.

3.4.2. Sustainability indicators

3.4.2.1. Socio-economic domain

Table 1

Candidate crops and their basic information and rotation rules.

Crop No.	Crop name	Botanical family	Planting date	Harvest date	Max. Freq. ^a	Min. Rep. ^b (yrs)
1	Winter wheat	Graminae	Oct.15	Jun.1	1	0
2	Summer maize _{normal}	Graminae	Jun.10	Oct.1	1	0
3	Summer maize _{late}	Graminae	Jul.5	Nov.15	1	0
4	Spring maize	Graminae	May.20	Oct.1	1	0
5	Foxtail millet _{normal} ^c	Graminae	Jun.5	Oct.1	0.5	1.5
6	Foxtail millet _{late} ^c	Graminae	Jul.5	Nov.15	0.5	1.5
7	Spring mung bean ^d	Fabaceae	Mar.31	Jul.1	0.5	1.5
8	Spring peanut ^d	Fabaceae	Apr.15	Aug.25	0.5	1.5
9	Cotton	Malvaceae	Apr.20	Oct.28	1	0
10	Stevia	Asteraceae	Apr.15	Aug.30	0.5	1.5
11	Chili pepper	Solanaceae	Apr.5	Sep.5	0.5	1.5
12	Cabbage	Cruciferae	Aug.1	Oct.1	0.5	1.5
13	Garlic	Amaryllidaceae	Oct.5	Jun.3	0.5	1.5

^a Max. Freq. means the maximum frequency of the given crop in a rotation.

^b Min. Rep. means the minimum period before repeated cultivation of the given crop in a rotation.

^c Max. Freq. and Min. Rep. for the crop group of foxtail millets are 0.5 and 1.5 years, respectively.

^d Max. Freq. and Min. Rep. for the crop group of legumes are 0.5 and 1.5 years, respectively.

- **Gross margin (GM, USD/ha/yr)** was calculated by dividing the sum of the gross margin of component crops of a rotation by the rotation length measured in years. To reduce the influence of price fluctuation, here we calculated the average gross margin of a rotation across 15 price settings corresponding to prices of harvested commodities and agricultural inputs over 15 years (2006–2020). Accounting for a 15-year period of price fluctuation aims to capture the full range of price fluctuation of agricultural commodities and materials, which are known to be highly volatile in Chinese and world markets (Headey and Fan, 2008).

$$GM = \frac{1}{15} \left(\sum_k GM_k \right) \quad (1)$$

$$GM_k = \left(\sum_i Y_{com,i} \times PR_{i,k} - C_{i,k} \right) / RL / CPI_k \quad (2)$$

Where k (=2006, 2007, ..., 2020) is the price setting index, i (=1, 2, ..., n) is the component crop, and n is the total number of component crops in a rotation. GM_k (USD/ha/yr) is the gross margin of a rotation in the price setting k . For crop i , $Y_{com,i}$ (Mg/ha) is the harvested commodity yield, and $PR_{i,k}$ (USD/Mg) is the market price of the harvested commodity in year k . $C_{i,k}$ (USD/ha) indicates the variable cost of crop i with input prices in year k , which includes costs of seeds or seedlings, plastic films, inorganic fertilizers, animal manures, electricity for irrigation, pesticides, and hired agricultural machinery. CPI_k is the consumer price index of year k , which was used to correct for the inflation across years so that the gross margins of different years were standardized to the year 2020. RL (years) denotes the rotation length. The price data and CPI across 15 price settings were obtained from the local market and statistical yearbook (HPBS, 2021). We assume the prices of crop input and output as exogenous so these would not vary with the prospective changes in crop rotation pattern at the county scale. The economic performance of candidate crops is shown in Table S4.

- **Gross margin variability (GMV, USD/ha/yr)** is the standard deviation of the gross margin of a rotation across 15 price settings (Smith et al., 2017):

$$GMV = SD(GM_k) \quad (3)$$

- **Labor requirement (L , h/ha/yr)** is the annual mean labor input required by the component crops of a rotation:

$$L = \sum_i L_i / RL \quad (4)$$

Where L_i (h/ha/yr) represents the labor requirement for crop i covering labor consumed for seeding or transplanting, fertilization, irrigation, crop protection, pruning, and harvesting.

3.4.2.2. Human-nutrition domain

- **Dietary energy yield, zinc yield, and vitamin C yield (Y_j , persons/ha/yr).** These three indicators reflect how many persons have their nutrient requirement met by crop products of a rotation in terms of dietary energy, zinc, and vitamin C respectively (DeFries et al., 2015). These were calculated by dividing the sum of the mass output of a given nutrient of all component crops in a rotation by the rotation length and associated recommended nutrient intake for a reference person:

$$Y_j = \sum_i Y_{com,i} \times F_{j,i} / RAI_j / RL \quad (5)$$

Where j (=Energy, Zinc, Vitamin C) is the index of the dietary nutrient. $F_{j,i}$ (Gcal or g/Mg) represents the fraction of nutrient j in the harvested commodity of crop i . RAI_j (Gcal or g/person/yr) means the recommended annual intake of nutrient j of a reference person (an 18-year-old male with a medium intensity of physical work). $F_{j,i}$ and RAI_j are listed in Table S5.

3.4.2.3. Eco-environment domain

- **Groundwater depletion (GWD, mm/yr)** was estimated by empirical equations (Yang et al., 2021):

$$GWD = \left[\left(\sum_i IR_i \right) - D \right] / RL \quad (6)$$

Where IR_i (mm) denotes the irrigation water amount used for crop i , and D (mm) means the deep infiltration during a rotation, calculated as:

$$D = \alpha \times \left[\left(\sum_i IR_i \right) + P \times RL \right] \quad (7)$$

Where P (mm/yr) is the annual precipitation in Quzhou. α represents the unitless recharge coefficient calibrated in the study area (0.2).

- **Eutrophication potential (EUP, kg PO₄ eq./ha/yr)** was used to quantify the life-cycle environmental influence of a rotation on the water bodies (ISO, 2006). The system boundary includes the pre-farm period, i.e., the production and transportation of agricultural inputs, and the on-farm period, i.e., fertilizer application on the arable field. The eutrophication potential of a rotation was calculated by using the below equation:

$$EUP = \sum_i (EUP_{pre,i} + EUP_{on,i}) / RL \quad (8)$$

Where $EUP_{pre,i}$ and $EUP_{on,i}$ (kg PO₄ eq./ha) denote the eutrophication potential of component crop i during the pre- and on-farm period, respectively. Detailed calculation of $EUP_{pre,i}$ and $EUP_{on,i}$ is specified in the section S1.1–1.2 of the Supplementary Materials.

- **Net GHG emission (GHG_{net} , kg CO₂ eq./ha/yr)** from the life-cycle perspective was used to assess the effects of a rotation on global warming (ISO, 2006):

$$GHG_{net} = \sum_i (GHG_{pre,i} + GHG_{on,i} - Q_{SOC}) / RL \quad (9)$$

Where $GHG_{pre,i}$ and $GHG_{on,i}$ (Mg CO₂ eq./ha) represent GHG emission during the pre- and on-farm period for crop i , respectively. Their calculation was specified in S1.1 and S1.3 of Supplementary Materials. Q_{SOC} (Mg CO₂ eq./ha) is the soil carbon stock change over a crop rotation cycle, estimated by ROTSOM model, which has been locally validated (see S1.3 of Supplementary Materials).

- **Pesticide input** is the annual mean of the amount of active ingredients (AI) of pesticides applied to all component crops in a rotation, expressed in kg AI/ha/yr. Pesticides in our study include insecticides, herbicides, and fungicides.

- **Agrobiodiversity (DIV, unitless).** Crop rotational diversity is highly associated with below- and above-ground biodiversity in

agroecosystems (Beillouin et al., 2021) so it can be a proxy indicator to quantify agrobiodiversity performance of a rotation. Referring to Socolar et al. (2021), we quantified crop rotational diversity by taking the geometric mean of species number and length of each rotation:

$$DIV = \sqrt{SN \times RL} \quad (10)$$

Where SN is the number of crop species with different ecological characteristics in a rotation. Wheat and foxtail millet were treated as species with the same characteristics as they have similar functional traits, e.g., they are both short-statured, deep-rooting, and tillering cereal crops.

3.5. Crop rotation selection

Selecting the outperforming rotations among a large set of generated rotations in terms of multiple indicators requires multi-attribute decision-making tools. In this study, we used two Pareto-based metrics to identify rotations that reached compromises across the entire indicator set.

3.5.1. Pareto rank-based selection

Pareto ranking is an efficient procedure to identify the better-performing individuals from a population of cases. Its major advantage is avoiding the application of weights, so an objective selection might be achieved (Adelhart Toorop et al., 2020). Here, we performed Pareto ranking using 11 sustainability indicators. In the ranking process, rotations were assigned rank “1” (i.e., Pareto-optimality) if none of the objectives was improved by another rotation without these other rotations being inferior in at least one of the other objectives (Groot et al., 2012). This process was conducted by *pareto_rank* function of “eaf” package in R. Here we kept only Pareto-rank-1 rotations for further selection.

3.5.2. Pareto efficiency-based selection

The power of Pareto ranking to narrow down the solution space becomes weaker when there are many indicators so that a large proportion of cases from a set would be identified as Pareto-optimal. More importantly, Pareto optimal cases include those that stand out in only one or a few indicators while performing unacceptably in others (Das, 1999). Given the above limitations, we also used Pareto efficiency score ordination to further select rotations with compromises across indicators from the Pareto-optimal set.

To do so, using the *combn* and *pareto_rank* function in R, we determined the Pareto efficiency score for each Pareto-optimal rotation according to the following principle: When the size of the whole indicator set is n , the Pareto efficiency score of a given case would be k ($1 \leq k \leq n$) if this case was Pareto-optimal for any of all possible k -element subsets of n indicators (Das, 1999). The lower the Pareto efficiency score is, the more compromise a rotation achieves across the indicator set (Fig. 2). We classified a rotation as a compromise rotation when its Pareto efficiency score was not higher than that of wheat-maize; that is: such compromise rotations had a trade-off across indicators that was at most as strong as that of the currently dominant wheat-maize system. We did not select only rotations with the lowest Pareto efficiency score, the number of which tends to be limited, because we deemed it necessary to maintain a diversity of solutions to meet different sustainability demands.

3.6. Crop rotation characterization

Systematically characterizing the sustainability profiles of the compromise rotations allows stakeholders to choose rotations that fit their sustainability demands. However, it can be expected that the performance of these promising rotations was heterogeneous. Therefore, we performed a multivariate clustering (Groot and Yang, 2022) to partition all compromise rotations into several clusters with contrasting sustainability patterns.

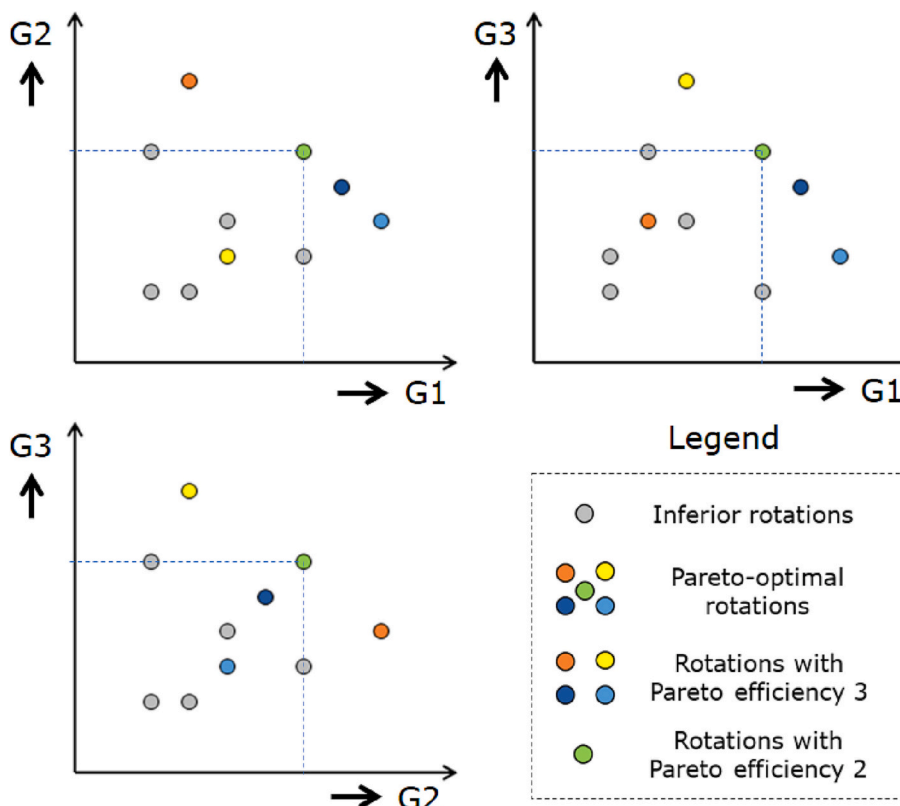


Fig. 2. Conceptual diagram of the determination of Pareto efficiency score for rotations. G1, G2, and G3 indicate three sustainability goals that are maximized. Circles filled with red, yellow, blue, purple, and green are Pareto-optimal rotations with respect to three goals (G1, G2, and G3), while grey circles are non-Pareto-optimal rotations, i.e., inferior rotations. Out of the five Pareto-optimal cases, only the green case is Pareto-optimal for all two-indicator subsets of G1, G2, and G3, which is therefore assigned a Pareto efficiency score of two, whereas the red, yellow, blue, and purple circles are assigned a Pareto efficiency score of three. The rotation with the lower Pareto efficiency score (2; green point) represents a better compromise across the entire indicator set.

This clustering method entailed a principal component analysis (PCA) and a hierarchical clustering analysis (HCA). With 11 sustainability indicators being response variables, we first conducted PCA using the *dudi.pca* function of the “ade4” R package to obtain principal components. Subsequently, we used HCA to classify all compromise rotations into different clusters in terms of principal components by using function *hclus* and *cutree* in R. The number of clusters was decided according to the bend position of the elbow plot (Kassambara, 2017). These rotation clusters served different patterns of sustainability performance.

Finally, we characterized the crop composition of compromise rotations in terms of the frequency (*Freq*) of four crop subsets, including cereals (wheat, maize, foxtail millet), legumes (peanut and mung bean), fiber crops (cotton) and cash crops (stevia, cabbage, chili pepper, and garlic). This frequency metric can be used to analyze the relations between crop choice and sustainability attributes of a rotation, which was calculated as follows:

$$Freq = N_s / RL \quad (11)$$

Where *s* is the index for crop subsets and *N_s* is the number of times crops from subset *s* were included in a rotation.

4. Results

4.1. Generated alternative rotations versus the currently dominant system

ROTAT generated 3011 feasible rotations with a length of 1–4 years for Quzhou, including the locally common wheat-maize double cropping system. Alternative rotations showed a wide range of performance in all indicators but none of them outperformed the wheat-maize system in all 11 indicators (Table 2). That is because the wheat-maize system had the highest dietary energy and zinc output. Furthermore, it had the fourth lowest labor requirement and a gross margin variability that was 48 % lower than the mean of alternatives. However, compared with the wheat-maize system, many alternative rotations attained large improvements in gross margin, vitamin C yield, and eco-environmental

Table 2

Indicator performance of the wheat-maize system and alternative rotations (*n* = 3010), and the number of rotations outperforming the wheat-maize system.

Indicator	Unit	Wheat-maize system	Alternative rotations ^a	Number of outperformers ^b
Gross margin	USD/ha/yr	2654	4446 (1641–8944)	2871
Gross margin variability	USD/ha/yr	392	755 (207–1677)	174
Labor requirement	h/ha/yr	247	1389 (189–3415)	3
Dietary energy yield	persons/ha/yr	57	28 (0–56)	0
Zinc yield	persons/ha/yr	69	36 (0–69)	0
Vitamin C yield	persons/ha/yr	0	184 (0–725)	2649
Groundwater depletion	mm/yr	284	223 (44–364)	2583
Eutrophication potential	kg PO ₄ eq./ha/yr	28.9	32.2 (12.3–60.0)	1012
Net GHG emission	Mg CO ₂ eq./ha/yr	6.64	7.52 (3.19–12.82)	850
Pesticide input	kg Al/ha/yr	6.64	9.77 (3.21–23.94)	895
Agrobiodiversity	unitless	1.41	4.16 (1.00–4.90)	3009

^a Data were shown as “mean (minimum–maximum)” across all alternative rotations.

^b Outperformers means the rotations that outperformed wheat-maize double cropping in the focus indicator.

indicators such as groundwater depletion, eutrophication potential, net GHG emission, and agrobiodiversity.

4.2. Pareto-optimal rotations and relationships among indicators

Among the 3011 generated rotations, 1298 rotations (43 %) were Pareto-optimal.

An indicator biplot matrix of the Pareto-optimal set shows relationships among sustainability indicators (Fig. 3). Within the socioeconomic domain, there were strong trade-offs between gross margin and gross margin variability or labor requirement. As for the human nutrition domain, dietary energy and zinc yield correlated positively, whereas vitamin C yield was highest at the intermediate levels of energy and zinc yield. In general, the eco-environmental indicators were positively associated with one another, while biodiversity tended to be highest for rotations with intermediate performance for other indicators. Pesticide use had a three-modal distribution within the set of indicators. This distribution was related to the cultivation frequency of chili pepper, the most pesticide-intensive crop among the candidate crops (Table S1).

When analyzing relationships among indicators in different sustainability domains, trade-offs mainly occurred in these pairs: gross margin versus dietary energy yield, gross margin versus environmental indicators, and nutritional yield versus environmental indicators. On the other hand, synergies existed between gross margin and vitamin C yield, and between gross margin variability or labor requirement and most environmental indicators.

4.3. Pareto efficiency-based selection: compromise rotations

Within the Pareto-optimal set of 1298 rotations, 1173 had higher Pareto efficiency score (10 or 11) than the currently dominant wheat-maize double cropping system which has a Pareto efficiency score of 9 (Table S6). The remaining 125 rotations, having a score of 9 or less, were selected as compromise rotations. This subset excluded extremely underperforming rotations in terms of gross margin variability, labor use, energy and zinc yield, and all eco-environmental indicators (Fig. 3). These compromise rotations were distributed widely across the indicator space (Fig. 3), indicating a great heterogeneity in indicator performance.

There were 109 rotations with a Pareto efficiency score of 9, equal to wheat-maize, but only sixteen with a Pareto efficiency score lower than 9 (Table S6): fourteen rotations with Pareto efficiency score of 8 and two rotations with Pareto efficiency score of 7. These sixteen alternatives had high frequencies of cereals (wheat, maize, and foxtail millet), legumes (peanut and mung bean), and garlic. The lowest score of 7 was attained by two four-year rotations: (Spring maize)-(Spring peanut)-(Spring maize)-(Spring mung bean)-(Foxtail millet_late) and (Spring maize)-(Spring peanut)-(Spring maize)-(Garlic)-(Foxtail millet_normal). While these rotations are well-performing in many respects, they did not meet all sustainability needs. For instance, compared with the wheat-maize system, these rotations resulted in 22–67 % lower groundwater depletion, eutrophication potential, GHG emission, and pesticide input but at the cost of 28 % lower gross margin for the former one, and a three times higher labor use for the latter one, and half dietary energy yield for both (Table S7).

4.4. Characterization of compromise rotations

Based on the PCA and HCA with 11 sustainability indicators, all 125 compromise rotations were divided into eight clusters (C1–C8; Fig. 4, Fig. S2).

C1 and C2 were characterized by a high frequency of cash crops. Rotations in these clusters delivered the greatest mean gross margin (4706–4748 USD/ha/yr) but resulted in the highest economic instability (981 USD/ha/yr) and labor requirement (1568 h/ha/yr), respectively. These profitable clusters generated relatively high environmental costs,

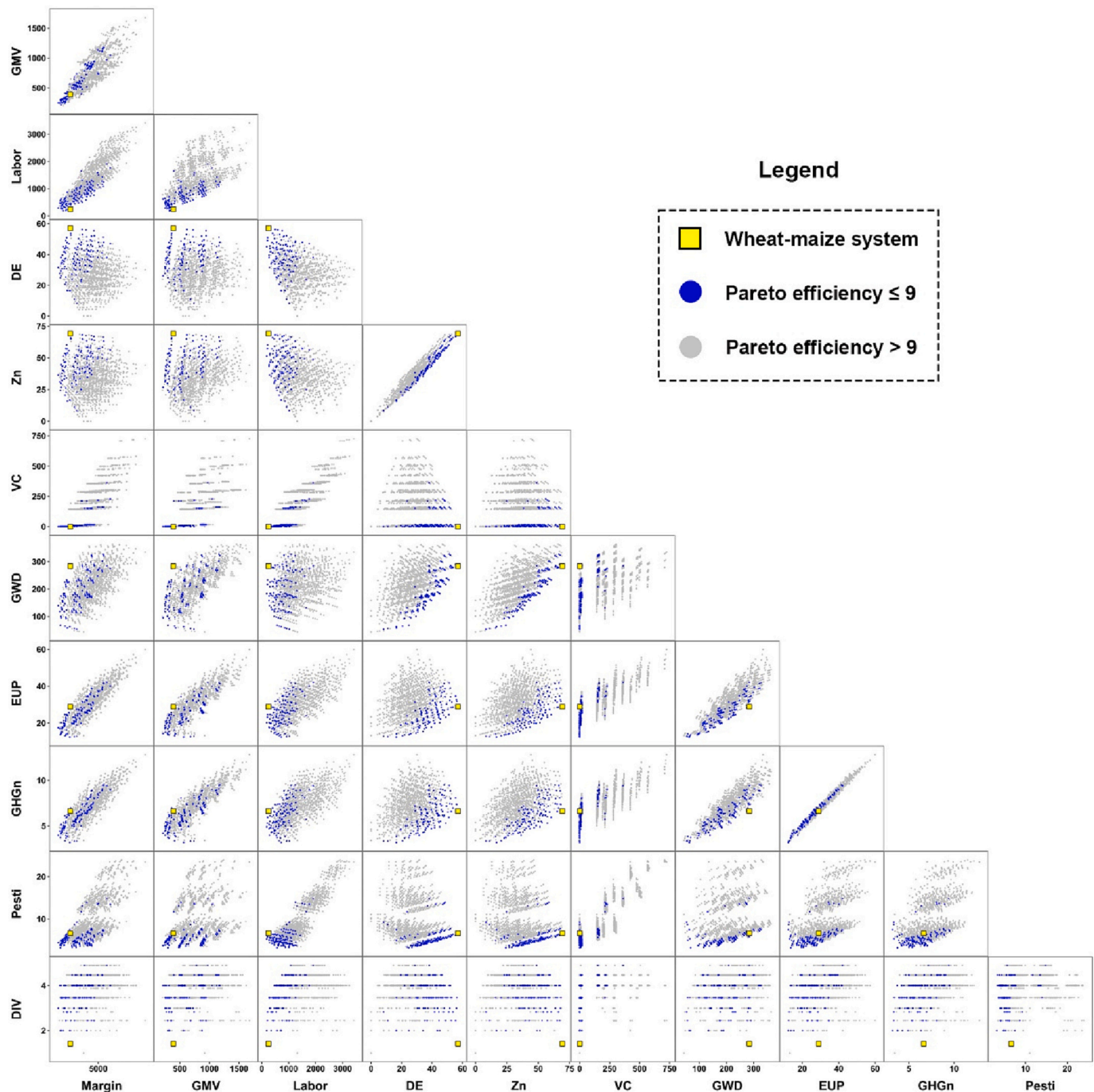


Fig. 3. Indicator biplot matrix for the entire set of Pareto-optimal rotations divided into two groups with Pareto efficiency score ≤ 9 (blue dots) and > 9 (grey dots). Rotations with Pareto efficiency score ≤ 9 were defined as compromise rotations. The yellow square denotes the currently dominant system (the wheat-maize system), for which Pareto efficiency score is 9. Margin, gross margin (USD/ha/yr); GMV, gross margin variability (USD/ha/yr); DE, dietary energy yield; Zn, zinc yield; VC, vitamin C yield (above three expressed in persons/ha/yr); GWD, groundwater depletion (mm/yr); Pesti, pesticide input (kg AI/ha/yr); EUP, eutrophication potential (kg PO₄ eq./ha/yr); GHGn, net greenhouse gas emission (Mg CO₂ eq./ha/yr); DIV, agrobiodiversity (unitless).

including groundwater depletion, eutrophication potential, and net GHG emission, while they supplied more dietary macro- and micro-nutrients and facilitated higher agrobiodiversity on average. Furthermore, C2 had the highest pesticide input.

C3 (which included the wheat-maize system) and C4 were clusters of cereal-dominated rotations. They were the most productive sets of rotations in terms of energy yield (49 persons/ha/yr) and zinc output (58 persons/ha/yr) across all clusters. On average, C3 had a low gross margin but with a low economic risk and labor use as well, whereas C4 obtained intermediate gross margin but at the cost of a higher economic

risk and labor use. Both C3 and C4 caused high groundwater depletion. The eutrophication potential and GHG emission of C4 were on average the third highest while those of C3 were intermediate among all clusters. C3 performed poorly in terms of agrobiodiversity (2.71).

Clusters C5 and C6 both had a high cultivation frequency of legumes and negligible vegetable production. C5 showed the highest frequency of cotton while C6 planted cereals instead of cotton. These two clusters had superior environmental performance. C5 resulted in the lowest groundwater depletion (65 mm/yr), eutrophication potential (13.0 kg PO₄ eq./ha/yr), and GHG emission (3.46 Mg CO₂ eq./ha/yr) while

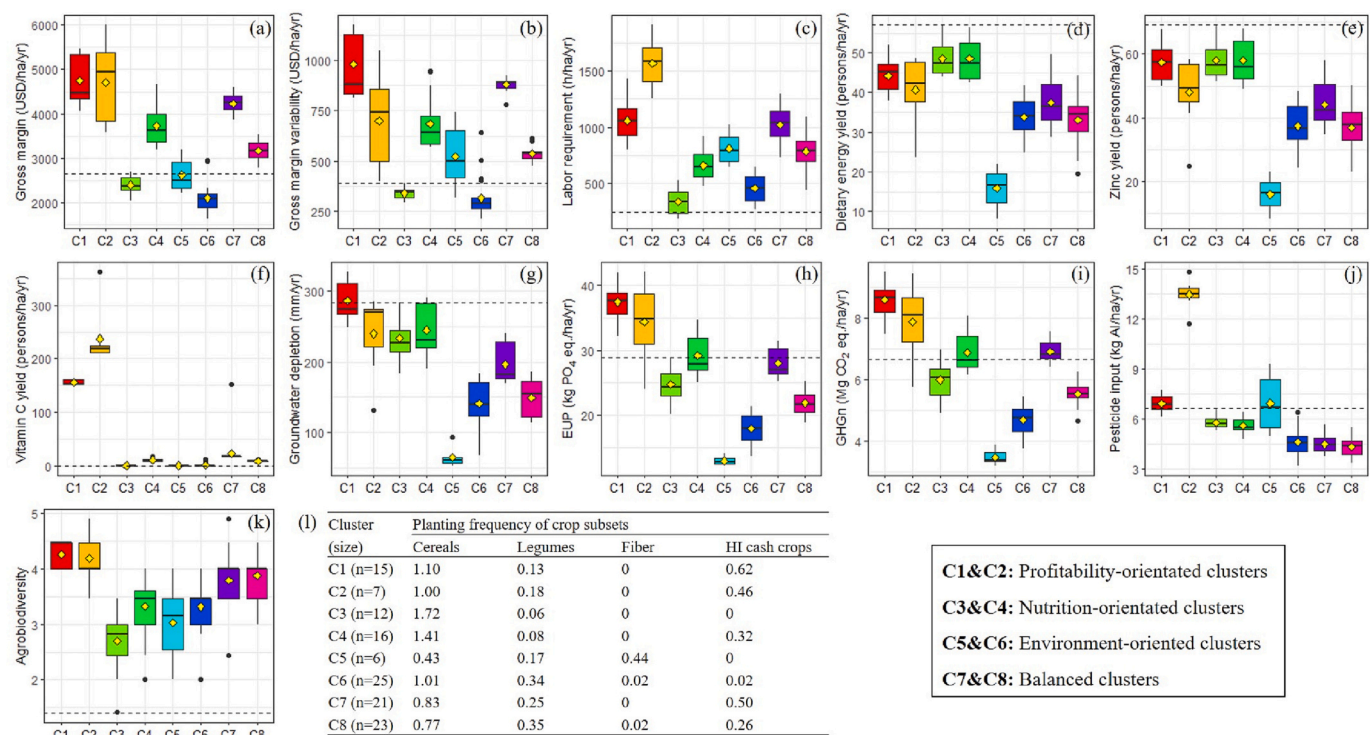


Fig. 4. Sustainability performance (a-k) and crop subset composition (l) of clusters (C1-C8) of compromise rotations. The yellow rhombus indicates the average value of a cluster, and the dashed line indicates the performance of the wheat-maize system. EUP means eutrophication potential and GHGn means net greenhouse gas emission. Data for planting frequency of crop subsets are the mean values of rotations in a cluster. “HI cash crops” means high-input cash crops. For members of each crop subset, see Section 3.6 in the main text.

markedly sacrificing economic and nutritional outputs. C6 was characterized by the second lowest groundwater depletion, eutrophication pollution, and GHG emission, as well as the third lowest pesticide use, but with the lowest gross margin.

C7 and C8 had high shares of legumes and intermediate shares of cereals and high-input cash crops. In contrast to the abovementioned clusters, these two clusters reached balanced performance across the indicator set. That means they performed at an intermediate level in nearly all indicators.

5. Discussion

In this study, we presented a methodological framework for systematic design of diversified crop rotations featuring compromises across multiple competing sustainability attributes. This is the first rotation design framework that combines a rotation generator model with Pareto-based decision-aid methods, to efficiently suggest options to advance sustainability. We illustrated this framework with data from a highly specialized and intensified cropping region (Quzhou county in the North China Plain) to explore promising rotations in terms of 11 sustainability indicators. Although there was no one-size-fits-all rotation that outperformed the locally dominant wheat-maize system for all indicators concurrently, this analysis identified a set of compromise rotations, i.e., rotations with trade-offs across indicators not stronger than that of the wheat-maize system. These compromise rotations also represented contrasting sustainability profiles, with different rotations satisfying to different extents objectives related to economic-value creation, human nutrition, and environmental conservation, or all of these in a balanced manner, depending on the crop species composition of the rotation. The diversity of compromise rotations could serve to meet diverse sustainability demands for stakeholders and complement the locally dominant cropping system in the future.

5.1. The future role and implementation of compromise rotations in specialized cropping regions

Intensified cereal-based systems in short crop rotations are widespread in many agricultural regions globally such as the North China Plain, the Indo-Gangetic Plain, and the Great Plain and Midwest of the US. Such systems are widely cultivated to maximize staple grain production with the supportive institutions and policies. For example, staple cereal production in China is promoted by policy instruments for the minimum procurement price, and grain and input subsidies (Kimura et al., 2019; Song et al., 2021; OECD, 2022), which provide a safety net for farmers. With the extensive cultivation of cereals, social institutions are increasingly prone to grain production accordingly. For instance, many agricultural markets and most technological innovations target staple cereals. Hence there is a socio-technical lock-in of cereal crop specialization (Meynard et al., 2018). However, cereal monoculture has rendered lots of socio-economic, human-nutrition, and eco-environmental issues so a cropping system redesign towards diversification is deemed necessary to enhance farming sustainability in these specialized cropping regions (Davis et al., 2019; Hunt et al., 2019; Yang et al., 2021). In China, the government has partially relaxed the food self-sufficiency policy (Ito and Li, 2023) and enacted various policies on Agriculture Green Development to foster multi-attribute agricultural sustainability (MARA, 2022a, 2022b) with a focus on the establishment of innovative cropping systems, which offer great opportunities for future application of diversified crop rotations.

As trade-offs among competing objectives are normally inevitable (Hegwood et al., 2022), promising cropping systems should reach compromises across sustainability domains. In our case study, with the help of the Pareto efficiency-based selection, our compromise rotations did not include extreme alternatives included in the Pareto-optimal set, for example, the most profitable system [a two-year rotation of (cabbage)-(chili pepper)-(garlic)] and the most groundwater-friendly system

(continuous cotton cropping), because these alternatives strongly sacrificed other sustainability indicators. We also found that compromise rotations comprised mainly wheat and maize supplemented with other candidate species, rather than rotations of pure vegetables or legumes. This implies that shifting from the wheat-maize system to these compromise rotations would not be a radical process, thus conducive to their acceptability to practitioners from the technical perspective.

Compromise rotations could be chosen and implemented tailored to the production contexts and stakeholders' interests because these rotations could deliver contrasting patterns of sustainability services according to multi-criteria characterization (Fig. 4; Fig. 5). Taking the North China Plain as an example, rotations from the profitability-oriented cluster (C1&C2) could be recommended to villages where boosting farmers' income is prioritized. These profitable rotations could act as a measure in policies on rural poverty reduction while policy-makers could provide accompanying measures, e.g., agricultural insurance, to hedge against market risks of vegetables (Feliciano, 2019). Stevia is now promoted in Quzhou to increase farm profits but is not included in any compromise rotation because it does not produce nutrients for human diets and has a high environmental burden, suggesting that non-edible and high-input cash crops might be not a reasonable choice when multiple objectives are considered. Nutrition-oriented rotations (C3&C4) could be practiced in villages that still shoulder the responsibility for food provisioning allocated by the Chinese government. Compared with wheat-maize, most rotations from C3 could reduce the environmental burden while those from C4 could enhance gross margin, at the cost of calorie production to some degree for both. The final decision can be made based on local stakeholders' preferences. The environmentally sound rotations can play a role in ecologically fragile villages with, for instance, severe groundwater overdrafts or nutrient pollution. Some of these rotations had a comparable or higher gross margin than wheat-maize owing to the inclusion of peanut and cotton, while others were less profitable thus requiring subsidies to incentive implementation. Rotations from balanced clusters are suitable

for areas without overriding priorities for specific sustainability domains. Relative to wheat-maize, these systems can improve gross margin and environmental performance with intermedium dietary nutrient supply by rotating cereals, legumes, and garlic. This result indicates that the balanced planting frequency of crops from different functional groups could balance trade-offs among conflicting sustainability indicators. Furthermore, as nearly all alternative rotations require more labor than wheat-maize, machinery for non-cereal crops could be devised and promoted to substantially reduce labor use thus facilitating the application of these compromise rotations. Taken together, through rational crop species arrangement, compromise rotations could act as a range of well-planned options to address different sustainability concerns in specialized cropping regions (Fig. 5; Table S8).

As mentioned before, trade-offs among indicators exist within clusters of rotations that had the same sustainability orientation, although softer than those among rotations with contrasting orientations. Therefore, stakeholder discussion might be needed to make a final choice for the rotations most suitable for adoption. The arable farming system involves several stakeholders, including crop farmers, governmental departments, and the public, with differing preferences for agricultural sustainability. Farmers primarily seek to maximize profits while minimizing market risks and labor inputs. On the other hand, the government and the public focus on securing food supply and reducing agriculture-induced resource and environmental costs. The set of compromise rotations could serve as a basis for evaluation and discussion among stakeholders, which would facilitate identifying the “middle ground” of competing interests while offering prioritized options for specific stakeholders' interests. Furthermore, as the compromise rotations were identified by inter-comparing all possible rotations, it is crucial to check and judge the specific quantities of each indicator of rotations during the stakeholders' decision-making process, allowing for better understanding of their potential benefits and drawbacks before implementation.

Improved management practices for candidate crops in crop

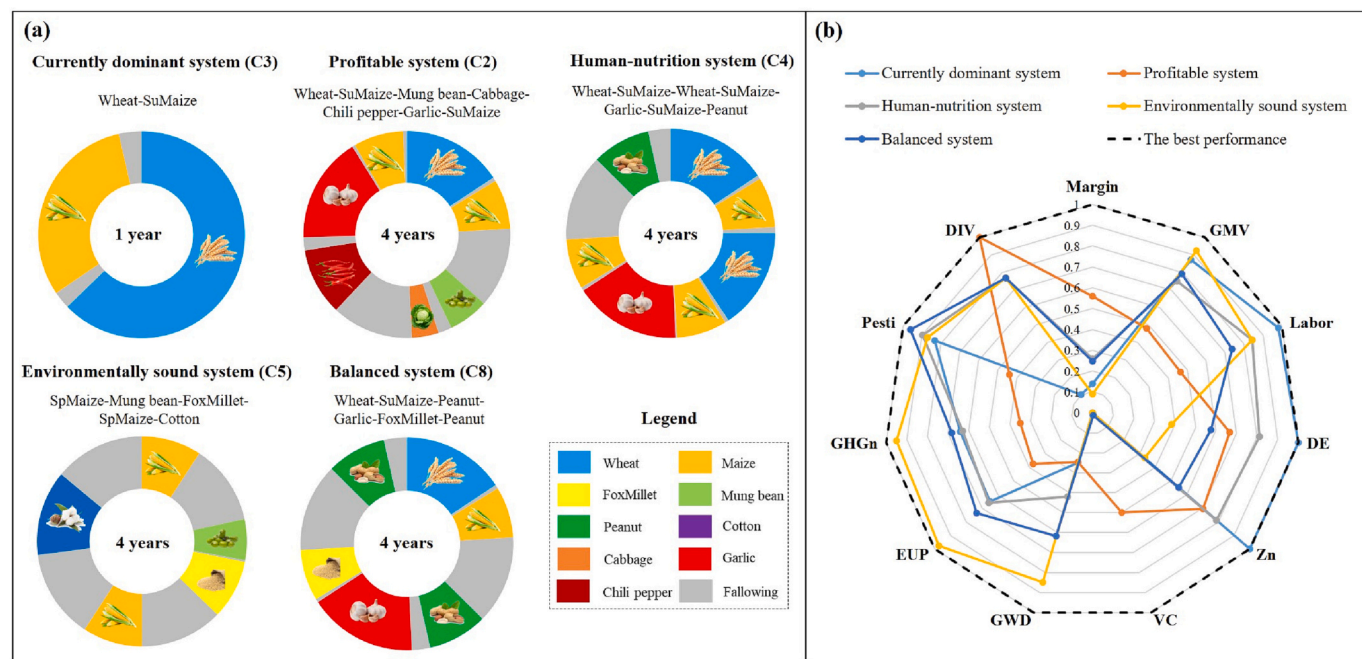


Fig. 5. The examples of compromise rotations (a) and comparison of their sustainability performance (b). Each donut plot represents a rotation, in which the length of each section indicates the proportion of crop growth and land fallowing period in the length of one rotation cycle. In subplot (b), values of each indicator were rescaled to 0–1 using Min-Max normalization. “0” and “1” correspond to the worst and best performance of each indicator across all generated rotations respectively, i.e., all indicators have the same optimization direction. Margin, gross margin; GMV, gross margin variability; DE, dietary energy yield; Zn, zinc yield; VC, vitamin C yield; GWD, groundwater depletion; Pesti, pesticide input; EUP, eutrophication potential; GHGn, net greenhouse gas emission; DIV, agrobiodiversity. Specific indicator values of each example rotation were listed in Table S8.

rotations could further ameliorate agricultural sustainability in the future. As we based our input-output relations on survey data, the farmer management for the candidate crops in this study may not be agronomically optimal. Further improvements might reduce yield gaps and resource inputs (Table S1; Table S9). If those crops are managed with efficient cultivation practices such as optimized nutrient and water management, the crop performance would accordingly improve, and, more importantly, crop interactions between consecutive crops in a rotation could perhaps function to additionally enhance system performance (Lambers and Cong, 2022), for instance, increased grain yield of cereals when preceded by N-fixing legumes in the rotation (Zhao et al., 2022; Xia et al., 2023).

5.2. Improved method for model-aided multi-objective crop rotation design

Model-aided rotation planning is a powerful approach in the field of multi-objective farming system redesign. Several studies used rule-based rotation generator models with multi-indicator assessment to offer a large set of all agronomically feasible alternative rotations and their performance but normally ended up analyzing the co-benefits or trade-offs among indicators or selecting preferred alternatives based on subjectively defined threshold values for some of the indicators (Dogliotti et al., 2004; Bachinger and Zander, 2007; Reckling et al., 2016). Here, our methodological framework introduces the Pareto-based multi-criteria decision-aid tools to select promising alternatives, where the performance features compromises in trade-off frontiers across objectives and is subsequently characterized by a clustering technique. Such clustering results could help stakeholders to choose alternatives fitting their demands in the follow-up decision-making process. Our framework integrates the advantages of the rotation generator and Pareto-based trade-off analysis, which can design rotations that are agronomically viable and better performing in terms of multiple attributes.

Our methodological framework suggests a set of promising rotations using Pareto-based metrics, rather than a single ‘optimal’ rotation using other decision-making methods such as aggregated sustainability indicators or data envelopment analysis (Cicciù et al., 2022). We think the method of rotation design should not aim to recommend a single solution for broader adoption but it should offer a range of solutions for different sustainability demands for a given agricultural region. The first reason is, as mentioned in Section 5.1, that trade-offs among sustainability indicators are normally unavoidable across solution space when a large number of indicators are considered (Hegwood et al., 2022). This makes it impossible to identify a solution that excels in all respects. It would be more practical to give a set of promising alternatives with an appropriate size to inform stakeholders’ decision-making based on their demands and preference. The other reason why a range of solutions should be offered is that promoting a single rotation across a region is not ecologically desirable. It can be expected that a wide application of a certain rotation, even if it is highly temporally diversified, would lead to homogenous agricultural landscapes, which could adversely affect the delivery of multiple ecosystem services (Sirami et al., 2019). Conversely, adoption of a range of alternative rotations within a region could help to construct more diversified landscapes.

The presented framework can be applied in other intensified and specialized cropping regions to advance agricultural sustainability with some adaptations. Firstly, sustainability indicators should be adjusted to align with region-specific objectives. For example, energy depletion for fossil fuel-scarce areas (Nemecek et al., 2015), soil acidification rate for acid-soil areas (Zhu et al., 2018), and soil erosion for the hilly areas (Dogliotti et al., 2004). The human-nutrition indicators can also be determined according to specific nutrient deficiency issues, e.g., selenium, iron, and vitamin A. Next, the candidate crop list and related input-output data for indicator assessment can be compiled based on local farmers’ survey, statistical materials, or expert knowledge. Using a survey dataset provides updated and detailed information on crop

production patterns within a given region and researchers can gain many insights on cropping system innovation during the interview with farmers. Furthermore, the rotation rules can be formulated with agronomists, advisors, or experienced farmers in the study region or extracted from relevant literature. By incorporating these adaptations, researchers or practitioners can effectively apply the framework to design promising diversified rotations elsewhere.

5.3. Limitation and outlook

The limitation of this study mainly lies in the uncertainty of the *ex-ante* sustainability assessment of generated rotations. First, the candidate crop dataset was compiled based on the farmer survey in 2020, so we did not consider the interannual variations in crop input and output, which might be affected by different weather conditions. Although having only one-year data, we found the temperature and precipitation in the year 2020 in Quzhou were close to average conditions so our dataset represents the baseline of local crop production. In the future, crop input and output data from several years could be collected and analyzed to increase the representativeness of the results. Second, the environmental impact assessment of candidate crops was based on numerous empirical models rather than field measurements. To make our results closer to reality, we chose widely-used models and used crop and environmental parameters that match the local situation (Supplementary Materials S1.1–1.3). In the economic assessment we took prices to be exogenous, which is defensible as the agricultural markets operate on a larger scale than the studied county while Quzhou is relatively well connected to those markets. However, when applying this methodological framework in regions with more localized markets, endogeneity of price variables may arise, particularly in relation to some region-specific crops or locally scarce agricultural inputs.

Future efforts could focus on field testing, regional planning, and implementing new scenarios. Some of the compromise rotations could be tested in the field experimental station or on farm. The rotations for testing are supposed to be from different rotation clusters, for example, those in Fig. 5. This would allow us to compare outcomes between the *ex-ante* and *ex-post* assessment, and help to adjust the parameters used in the design approach and gain more insight in system functioning (Cong et al., 2021). Next, we could investigate how those promising alternative rotations would be spatially allocated to cropping landscape within the study region to pursue agricultural sustainability at the regional scale. The allocation needs to consider the potential biophysical and socio-economic barriers for adoption of those alternatives in a given land parcel and could account for interactions with surrounding fields and other landscape elements. Lastly, we could also explore alternative rotations towards better farming sustainability with new scenarios concerning, for instance, improvement in management practices and inclusion of locally underutilized crop species.

6. Conclusion

This study presented a new crop rotation design framework aiming to enhance multi-attribute sustainability for intensified and specialized cropping regions. The framework consists of a rotation generator model, Pareto-based decision-aid tools, and a multi-criteria characterization process, able to offer a set of diversified rotations with compromises across multiple sustainability attributes. We illustrated this framework with data from Quzhou county in the North China Plain, a region with double cropping of cereals as the dominant land use, and successfully proposed a manageable size of compromise rotations in the end. The currently dominant cropping system, wheat-maize double cropping, contributed most to dietary energy provisioning while the compromise alternatives improved gross margin, vitamin C production, and a range of environmental indicators. The resulting rotations were grouped into multiple clusters with contrasting sustainability profiles fitting corresponding production contexts and stakeholders’ preferences. Thus, with

this framework, we provided a proper diversity of solutions that might become important elements of the future farming landscape to support stakeholders' decision-making. With adaptations to contrasting conditions, this rotation design framework could also be applied in other specialized and intensified agricultural areas to combat sustainability challenges.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Adelhart Toorop, R., Ceccarelli, V., Bijarniya, D., Jat, M.L., Jat, R.K., Lopez-Ridaura, S., Groot, J.C.J., 2020. Using a positive deviance approach to inform farming systems redesign: A case study from Bihar, India. *Agric. Syst.* 185, 102942. <https://doi.org/10.1016/j.agsy.2020.102942>.
- Ankathi, S.K., Long, D.S., Gollany, H.T., Das, P., Shonnard, D., 2019. Life cycle assessment of oilseed crops produced in rotation with dryland cereals in the inland Pacific Northwest. *Int. J. Life Cycle Assess.* 24, 627–641. <https://doi.org/10.1007/s11367-018-1488-y>.
- Bachinger, J., Zander, P., 2007. ROTOR, a tool for generating and evaluating crop rotations for organic farming systems. *Eur. J. Agron.* 26, 130–143. <https://doi.org/10.1016/j.eja.2006.09.002>.
- Beckmann, M., Gerstner, K., Akin-Fajiyi, M., Ceaşu, S., Kambach, S., Kinlock, N.L., Phillips, H.R.P., Verhagen, W., Gurevitch, J., Klotz, S., Newbold, T., Verburg, P.H., Winter, M., Seppelt, R., 2019. Conventional land-use intensification reduces species richness and increases production: A global meta-analysis. *Glob. Chang. Biol.* 25, 1941–1956. <https://doi.org/10.1111/gcb.14606>.
- Beillouin, D., Ben-Ari, T., Malézieux, E., Seufert, V., Makowski, D., 2021. Positive but variable effects of crop diversification on biodiversity and ecosystem services. *Glob. Chang. Biol.* 27, 4697–4710. <https://doi.org/10.1111/gcb.15747>.
- Ben Abdallah, S., Parra-López, C., Elfik, S., Suárez-Rey, E.M., Romero-Gómez, M., 2022. Sustainability assessment of traditional, intensive and highly-intensive olive growing systems in Tunisia by integrating Life Cycle and Multicriteria Decision analyses. *Sustain. Prod. Consum.* 33, 73–87. <https://doi.org/10.1016/j.spc.2022.06.022>.
- Bennetzen, E.H., Smith, P., Porter, J.R., 2016. Agricultural production and greenhouse gas emissions from world regions—The major trends over 40 years. *Glob. Environ. Chang.* 37, 43–55. <https://doi.org/10.1016/j.gloenvcha.2015.12.004>.
- Carof, M., Colomb, B., Aveline, A., 2013. A guide for choosing the most appropriate method for multi-criteria assessment of agricultural systems according to decision-makers' expectations. *Agric. Syst.* 115, 51–62. <https://doi.org/10.1016/j.agsy.2012.09.011>.
- Cicciù, B., Schramm, F., Schramm, V.B., 2022. Multi-criteria decision making/aid methods for assessing agricultural sustainability: A literature review. *Environ. Sci. Pol.* 138, 85–96. <https://doi.org/10.1016/j.envsci.2022.09.020>.
- Cong, W.-F., Zhang, C., Li, C., Wang, G., Zhang, F., 2021. Designing diversified cropping systems in China: Theory, approaches and implementation. *Front. Agric. Sci. Eng.* 8 (3), 362–372. <https://doi.org/10.15302/J-FASE-2021392>.
- Das, I., 1999. A preference ordering among various Pareto optimal alternatives. *Struct. Optim.* 18, 30–35. <https://doi.org/10.1007/BF01210689>.
- Davis, K.F., Chhatre, A., Rao, N.D., Singh, D., Ghosh-Jerath, S., Mridul, A., Poblete-Cazenave, M., Pradhan, N., DeFries, R., 2019. Assessing the sustainability of post-green Revolution cereals in India. *Proc. Natl. Acad. Sci.* 116, 25034–25041. <https://doi.org/10.1073/pnas.1910935116>.
- DeFries, R., Fanzo, J., Remans, R., Palm, C., Wood, S., Anderman, T.L., 2015. Metrics for land-scarce agriculture. *Science* 349, 238–240. <https://doi.org/10.1126/science.aaa5766>.
- Deytieu, V., Munier-Jolain, N., Caneill, J., 2016. Assessing the sustainability of cropping systems in single- and multi-site studies. A review of methods. *Eur. J. Agron.* 72, 107–126. <https://doi.org/10.1016/j.eja.2015.10.005>.
- Dogliotti, S., Rossing, W.A.H., van Ittersum, M.K., 2003. ROTAT, a tool for systematically generating crop rotations. *Eur. J. Agron.* 19, 239–250. [https://doi.org/10.1016/S1161-0301\(02\)00047-3](https://doi.org/10.1016/S1161-0301(02)00047-3).
- Dogliotti, S., Rossing, W.A.H., van Ittersum, M.K., 2004. Systematic design and evaluation of crop rotations enhancing soil conservation, soil fertility and farm income: a case study for vegetable farms in South Uruguay. *Agric. Syst.* 80, 277–302. <https://doi.org/10.1016/j.agsy.2003.08.001>.
- Dogliotti, S., García, M.C., Peluffo, S., Dieste, J.P., Pedemonte, A.J., Bacigalupe, G.F., Scarlato, M., Alliaume, F., Alvarez, J., Chiappe, M., Rossing, W.A.H., 2014. Co-innovation of family farm systems: A systems approach to sustainable agriculture. *Agric. Syst.* 126, 76–86. <https://doi.org/10.1016/j.agsy.2013.02.009>.
- Dury, J., Schaller, N., Garcia, F., Reynaud, A., Bergez, J.E., 2012. Models to support cropping plan and crop rotation decisions. A review. *Agron. Sustain. Dev.* 32, 567–580. <https://doi.org/10.1007/s13593-011-0037-x>.
- Dwivedi, S.L., Lammerts van Bueren, E.T., Ceccarelli, S., Grando, S., Upadhyaya, H.D., Ortiz, R., 2017. Diversifying food systems in the pursuit of sustainable food production and healthy diets. *Trends Plant Sci.* 22, 842–856. <https://doi.org/10.1016/j.tplants.2017.06.011>.
- Feliciano, D., 2019. A review on the contribution of crop diversification to Sustainable Development Goal 1 “No poverty” in different world regions. *Sustain. Dev.* 27, 795–808. <https://doi.org/10.1002/sd.1923>.
- Foley, J.A., Ramankutty, N., Brauman, K.A., Cassidy, E.S., Gerber, J.S., Johnston, M., Mueller, N.D., O'Connell, C., Ray, D.K., West, P.C., Balzer, C., Bennett, E.M., Carpenter, S.R., Hill, J., Monfreda, C., Polasky, S., Rockström, J., Sheehan, J., Siebert, S., Tilman, D., Zaks, D.P.M., 2011. Solutions for a cultivated planet. *Nature* 478, 337–342. <https://doi.org/10.1038/nature10452>.
- Garibaldi, L.A., Gemmill-Herren, B., D'Annolfo, R., Graeb, B.E., Cunningham, S.A., Breeze, T.D., 2017. Farming approaches for greater biodiversity, livelihoods, and food security. *Trends Ecol. Evol.* 32, 68–80. <https://doi.org/10.1016/j.tree.2016.10.001>.
- Groot, J.C.J., Yang, X., 2022. Trade-offs in the design of sustainable cropping systems at a regional level: a case study on the North China Plain. *Front. Agr. Sci. Eng.* 9 (2), 295–308. <https://doi.org/10.15302/J-FASE-2021434>.
- Groot, J.C.J., Jellema, A., Rossing, W.A.H., 2010. Designing a hedgerow network in a multifunctional agricultural landscape: Balancing trade-offs among ecological quality, landscape character and implementation costs. *Eur. J. Agron.* 32, 112–119. <https://doi.org/10.1016/j.eja.2009.07.002>.
- Groot, J.C.J., Oomen, G.J.M., Rossing, W.A.H., 2012. Multi-objective optimization and design of farming systems. *Agric. Syst.* 110, 63–77. <https://doi.org/10.1016/j.agsy.2012.03.012>.
- Headey, D., Fan, S., 2008. Anatomy of a crisis: the causes and consequences of surging food prices. *Agric. Econ.* 39, 375–391. <https://doi.org/10.1111/j.1574-0862.2008.00345.x>.
- Hegwood, M., Langendorf, R.E., Burgess, M.G., 2022. Why win–wins are rare in complex environmental management. *Nat. Sustain.* 5, 674–680. <https://doi.org/10.1038/s41893-022-00866-z>.
- Hernández-Ochoa, I.M., Gaiser, T., Kersebaum, K.-C., Webber, H., Seidel, S.J., Grahmann, K., Ewert, F., 2022. Model-based design of crop diversification through new field arrangements in spatially heterogeneous landscapes. A review. *Agron. Sustain. Dev.* 42, 74. <https://doi.org/10.1007/s13593-022-00805-4>.
- HPBS (Hebei Provincial Bureau of Statistics), 2021. *Hebei Rural Statistical Yearbook*. China Statistics Press, Beijing (in Chinese).
- Hunt, N.D., Hill, J.D., Liebman, M., 2019. Cropping system diversity effects on nutrient discharge, soil erosion, and agronomic performance. *Environ. Sci. Technol.* 53, 1344–1352. <https://doi.org/10.1021/acs.est.8b02193>.
- Ickowitz, A., Powell, B., Rowland, D., Jones, A., Sunderland, T., 2019. Agricultural intensification, dietary diversity, and markets in the global food security narrative. *Glob. Food Secur.* 20, 9–16. <https://doi.org/10.1016/j.gfs.2018.11.002>.
- ISO (International Organization for Standardization), 2006. *Environmental management life cycle assessment: requirements and guidelines*. ISO Geneva, Switzerland.
- Ito, J., Li, X., 2023. Interplay between China's grain self-sufficiency policy shifts and interregional, intertemporal productivity differences. *Food Policy* 117, 102446. <https://doi.org/10.1016/j.foodpol.2023.102446>.
- Kanter, D.R., Musumba, M., Wood, S.L.R., Palm, C., Antle, J., Balvanera, P., Dale, V.H., Havlik, P., Kline, K.L., Scholes, R.J., Thornton, P., Titttonell, P., Andelman, S., 2018. Evaluating agricultural trade-offs in the age of sustainable development. *Agric. Syst.* 163, 73–88. <https://doi.org/10.1016/j.agsy.2016.09.010>.
- Kassambara, A., 2017. *Practical Guide to Cluster Analysis in R*. Unsupervised Machine Learning (Edition 1). STHDA. <http://www.sthda.com>.
- Kimura, S., Gay, S.H., Yu, W., 2019. China's grains policy: impacts of alternative reform options. In: *OECD Food, Agriculture and Fisheries Papers*. No. 129. OECD Publishing, Paris. <https://doi.org/10.1787/aed5174b-en>.
- Lambers, H., Cong, W.-F., 2022. Challenges providing multiple ecosystem benefits for sustainable managed systems. *Front. Agric. Sci. Eng.* 9 (2), 170–176. <https://doi.org/10.15302/J-FASE-2022444>.
- Li, J., Lei, H., 2021. Tracking the spatio-temporal change of planting area of winter wheat–summer maize cropping system in the North China Plain during 2001–2018. *Comput. Electron. Agr.* 187, 106222. <https://doi.org/10.1016/j.compag.2021.106222>.
- Li, K., Wang, C., Zhang, H., Zhang, J., Jiang, R., Feng, G., Liu, X., Zuo, Y., Yuan, H., Zhang, C., Gai, J., Tian, J., Li, H., Sun, Y., Yu, B., 2022. Evaluating the effects of agricultural inputs on the soil quality of smallholdings using improved indices. *CATENA* 209, 105838. <https://doi.org/10.1016/j.catena.2021.105838>.
- Liang, Z., van der Werf, W., Xu, Z., Cheng, J., Wang, C., Cong, W.-F., Zhang, C., Zhang, F., Groot, J.C.J., 2022. Identifying exemplary sustainable cropping systems using a positive deviance approach: Wheat–maize double cropping in the North China Plain. *Agric. Syst.* 201, 103471. <https://doi.org/10.1016/j.agsy.2022.103471>.
- Liu, C., Plaza-Bonilla, D., Coulter, J.A., Kutcher, H.R., Beckie, H.J., Wang, L., Floch, J.-B., Hamel, C., Siddique, K.H.M., Li, L., Gan, Y., 2022. Diversifying crop rotations

- enhances agroecosystem services and resilience. In: *Advances in Agronomy*. Elsevier. <https://doi.org/10.1016/bs.agron.2022.02.007> (p. S0065211322000347).
- MacLaren, C., Mead, A., van Balen, D., Claessens, L., Etana, A., de Haan, J., Haagsma, W., Jäck, O., Keller, T., Labuschagne, J., Myrbeck, Å., Necpalova, M., Nziguheba, G., Six, J., Strauss, J., Swanepoel, P.A., Thierfelder, C., Topp, C., Tshuma, F., Versteegen, H., Walker, R., Watson, C., Wesselink, M., Storkey, J., 2022. Long-term evidence for ecological intensification as a pathway to sustainable agriculture. *Nat. Sustain.* <https://doi.org/10.1038/s41893-022-00911-x>.
- MARA (Ministry of Agriculture and Rural Affairs of the People's Republic of China), 2022a. Implementation Plan for Establishing National Agricultural Green Development Pilot Areas and Promoting Comprehensive Green Transformation in Agricultural Modernization Demonstration Areas (In Chinese).
- MARA (Ministry of Agriculture and Rural Affairs of the People's Republic of China), 2022b. The Plan of National Crop Production Development for the 14th Five-Year (In Chinese).
- Meynard, J.-M., Charrier, F., Fares, M., Le Bail, M., Magrini, M.-B., Charlier, A., Messéan, A., 2018. Socio-technical lock-in hinders crop diversification in France. *Agron. Sustain. Dev.* 38, 54. <https://doi.org/10.1007/s13593-018-0535-1>.
- Mu, H., Wang, K., Yang, X., Xu, W., Liu, X., Ritsema, C.J., Geissen, V., 2022. Pesticide usage practices and the exposure risk to pollinators: A case study in the North China Plain. *Ecotoxicol. Environ. Saf.* 241, 113713 <https://doi.org/10.1016/j.ecoenv.2022.113713>.
- Nemecek, T., Hayer, F., Bonnin, E., Carrouée, B., Schneider, A., Vivier, C., 2015. Designing eco-efficient crop rotations using life cycle assessment of crop combinations. *Eur. J. Agron.* 65, 40–51. <https://doi.org/10.1016/j.eja.2015.01.005>.
- OECD, 2022. Agricultural Policy Monitoring and Evaluation 2022: Reforming Agricultural Policies for Climate Change Mitigation. OECD Publishing, Paris. <https://doi.org/10.1787/7f4542bf-en>.
- Ouda, S., Zohry, A.E.-H., Noreldin, T., 2018. Crop Rotation: An Approach to Secure Future Food. Springer International Publishing, Cham. <https://doi.org/10.1007/978-3-030-05351-2>.
- Pei, H., Scanlon, B.R., Shen, Y., Reedy, R.C., Long, D., Liu, C., 2015. Impacts of varying agricultural intensification on crop yield and groundwater resources: comparison of the North China Plain and US High Plains. *Environ. Res. Lett.* 10, 044013 <https://doi.org/10.1088/1748-9326/10/4/044013>.
- Pellegrini, P., Fernández, R.J., 2018. Crop intensification, land use, and on-farm energy-use efficiency during the worldwide spread of the green revolution. *Proc. Natl. Acad. Sci.* 115, 2335–2340. <https://doi.org/10.1073/pnas.1717072115>.
- Pokhrel, A., Soni, P., 2017. Performance analysis of different rice-based cropping systems in tropical region of Nepal. *J. Environ. Manag.* 197, 70–79. <https://doi.org/10.1016/j.jenvman.2017.03.035>.
- Reckling, M., Hecker, J.-M., Bergkvist, G., Watson, C.A., Zander, P., Schläpke, N., Stoddard, F.L., Eory, V., Topp, C.F.E., Maire, J., Bachinger, J., 2016. A cropping system assessment framework—Evaluating effects of introducing legumes into crop rotations. *Eur. J. Agron.* 76, 186–197. <https://doi.org/10.1016/j.eja.2015.11.005>.
- Rosa-Schleich, J., Loos, J., Mußhoff, O., Tscharnkte, T., 2019. Ecological-economic trade-offs of diversified farming systems – a review. *Ecol. Econ.* 160, 251–263. <https://doi.org/10.1016/j.ecolecon.2019.03.002>.
- Saber, Z., Van Zelm, R., Pirdashti, H., Schipper, A.M., Esmaeili, M., Motevali, A., Nabavi-Pelesaraei, A., Huijbregts, M.A.J., 2021. Understanding farm-level differences in environmental impact and eco-efficiency: The case of rice production in Iran. *Sustain. Prod. Consum.* 27, 1021–1029. <https://doi.org/10.1016/j.spc.2021.02.033>.
- Sánchez, A.C., Kamau, H.N., Grazioli, F., Jones, S.K., 2022. Financial profitability of diversified farming systems: A global meta-analysis. *Ecol. Econ.* 201, 107595 <https://doi.org/10.1016/j.ecolecon.2022.107595>.
- Sietz, D., Klimek, S., Dauber, J., 2022. Tailored pathways toward revived farmland biodiversity can inspire agroecological action and policy to transform agriculture. *Commun. Earth Environ.* 3, 211. <https://doi.org/10.1038/s43247-022-00527-1>.
- Sirami, C., Gross, N., Baillod, A.B., Bertrand, C., Carrié, R., Hass, A., Henckel, L., Miguet, P., Vuillot, C., Alignier, A., Girard, J., Batáry, P., Clough, Y., Violle, C., Giral, D., Bota, G., Badenhaußer, I., Lefebvre, G., Gauffre, B., Vialatte, A., Calatayud, F., Gil-Tena, A., Tischendorf, L., Mitchell, S., Lindsay, K., Georges, R., Hilaire, S., Recasens, J., Solé-Senan, X.O., Robleño, I., Bosch, J., Barrientos, J.A., Ricarte, A., Marcos-García, M.Á., Miñano, J., Mathevet, R., Gibon, A., Baudry, J., Balent, G., Poulin, B., Burel, F., Tscharnkte, T., Bretagnolle, V., Siriwardena, G., Quin, A., Brotons, L., Martin, J.-L., Fahrig, L., 2019. Increasing crop heterogeneity enhances multitrophic diversity across agricultural regions. *Proc. Natl. Acad. Sci.* 116, 16442–16447. <https://doi.org/10.1073/pnas.1906419116>.
- Smith, A., Snapp, S., Chikowo, R., Thorne, P., Bekunda, M., Glover, J., 2017. Measuring sustainable intensification in smallholder agroecosystems: A review. *Glob. Food Secur.* 12, 127–138. <https://doi.org/10.1016/j.gfs.2016.11.002>.
- Socolar, Y., Goldstein, B.R., de Valpine, P., Bowles, T.M., 2021. Biophysical and policy factors predict simplified crop rotations in the US Midwest. *Environ. Res. Lett.* 16, 054045 <https://doi.org/10.1088/1748-9326/abf9ca>.
- Song, X., Wang, X., Li, X., Zhang, W., Scheffran, J., 2021. Policy-oriented versus market-induced: Factors influencing crop diversity across China. *Ecol. Econ.* 190, 107184 <https://doi.org/10.1016/j.ecolecon.2021.107184>.
- Talukder, B., Hipel, K.W., vanLoon, G.W., 2018. Using multi-criteria decision analysis for assessing sustainability of agricultural systems. *Sustain. Dev.* 26, 781–799. <https://doi.org/10.1002/sd.1848>.
- Uluhan, D., Grillot, M., Benoit, M., Bernes, G., Dumont, B., Magne, M.-A., Monteiro, L., Parsons, D., Veyset, P., Ryschawy, J., Steinmetz, L., Martin, G., 2022. Positive deviant strategies implemented by organic multi-species livestock farms in Europe. *Agric. Syst.* 201, 103453 <https://doi.org/10.1016/j.agsy.2022.103453>.
- Wang, S., Hu, Y., Yuan, R., Feng, W., Pan, Y., Yang, Y., 2019. Ensuring water security, food security, and clean water in the North China Plain – conflicting strategies. *Curr. Opin. Environ. Sustain.* 40, 63–71. <https://doi.org/10.1016/j.cosust.2019.09.008>.
- Withers, P., Neal, C., Jarvie, H., Doody, D., 2014. Agriculture and eutrophication: where do we go from here? *Sustainability* 6, 5853–5875. <https://doi.org/10.3390/su6095853>.
- Xia, H., Qiao, Y., Li, X., Xue, Y., Wang, N., Yan, W., Xue, Y., Cui, Z., van der Werf, W., 2023. Moderation of nitrogen input and integration of legumes via intercropping enable sustainable intensification of wheat-maize double cropping in the North China Plain: A four-year rotation study. *Agric. Syst.* 204, 103540 <https://doi.org/10.1016/j.agsy.2022.103540>.
- Xian, Y., Cai, G., Lin, J., Chen, Y., Wang, X., 2023. Comparison of crop productivity, economic benefit and environmental footprints among diversified multi-cropping systems in South China. *Sci. Total Environ.* 874, 162407 <https://doi.org/10.1016/j.scitotenv.2023.162407>.
- Xiao, H., van Es, H.M., Amsili, J.P., Shi, Q., Sun, J., Chen, Y., Sui, P., 2022. Lowering soil greenhouse gas emissions without sacrificing yields by increasing crop rotation diversity in the North China Plain. *Field Crop Res.* 276, 108366 <https://doi.org/10.1016/j.fcr.2021.108366>.
- Yang, X., Steenhuis, T.S., Davis, K.F., Werf, W., Ritsema, C.J., Pacenka, S., Zhang, F., Siddique, K.H.M., Du, T., 2021. Diversified crop rotations enhance groundwater and economic sustainability of food production. *Food Energy Secur.* 10 <https://doi.org/10.1002/fes.3.311>.
- Yang, L., Zhou, Y., Meng, B., Li, H., Zhan, J., Xiong, H., Zhao, H., Cong, W., Wang, X., Zhang, W., Lakshmanan, P., Deng, Y., Shi, X., Chen, X., Zhang, F., 2022. Reconciling productivity, profitability and sustainability of small-holder sugarcane farms: A combined life cycle and data envelopment analysis. *Agric. Syst.* 199, 103392 <https://doi.org/10.1016/j.agsy.2022.103392>.
- Zhang, W., Cao, G., Li, X., Zhang, H., Wang, C., Liu, Q., Chen, X., Cui, Z., Shen, J., Jiang, R., Mi, G., Miao, Y., Zhang, F., Dou, S., 2016. Closing yield gaps in China by empowering smallholder farmers. *Nature* 537, 671–674. <https://doi.org/10.1038/nature19368>.
- Zhao, J., Yang, Y., Zhang, K., Jeong, J., Zeng, Z., Zang, H., 2020. Does crop rotation yield more in China? A meta-analysis. *Field Crop Res.* 245, 107659 <https://doi.org/10.1016/j.fcr.2019.107659>.
- Zhao, Y., Wang, L., Lei, X., Wang, B., Cui, J., Xu, Y., Chen, Y., Sui, P., 2021. Reducing carbon footprint without compromising grain security through relaxing cropping rotation system in the North China Plain. *J. Clean. Prod.* 318, 128465 <https://doi.org/10.1016/j.jclepro.2021.128465>.
- Zhao, J., Chen, J., Beillouin, D., Lambers, H., Yang, Y., Smith, P., Zeng, Z., Olesen, J.E., Zang, H., 2022. Global systematic review with meta-analysis reveals yield advantage of legume-based rotations and its drivers. *Nat. Commun.* 13, 4926. <https://doi.org/10.1038/s41467-022-32464-0>.
- Zhu, Q., De Vries, W., Liu, X., Hao, T., Zeng, M., Shen, J., Zhang, F., 2018. Enhanced acidification in Chinese croplands as derived from element budgets in the period 1980–2010. *Sci. Total Environ.* 618, 1497–1505. <https://doi.org/10.1016/j.scitotenv.2017.09.289>.